Planning and design for tunnels and caverns

The following are usually carried out at each stage of a tunnel/cavern development:

(a) *Planning* - Desk studies and site reconnaissance including interpretation of aerial photographs and use of remote sensing data.

(b) *Preliminary design* - Detailed engineering geological mapping(including discontinuity surveys), geophysical investigation, drillholes and initial groundwater investigation.

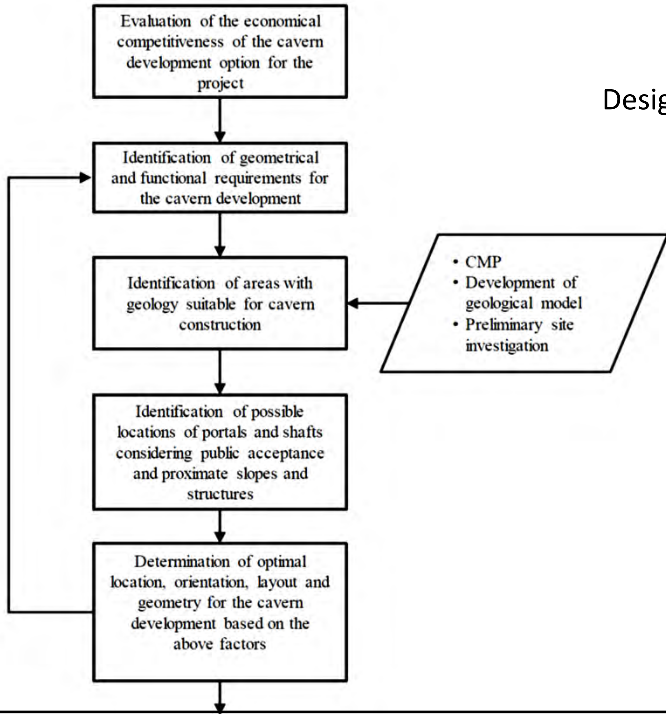
(c) *Detailed design* - Additional drillholes (likely including inclined and/or directional drillholes), geophysical investigation, detailed groundwater investigation and additional engineering geological mapping as necessary.

(d) *Construction* - Engineering geological mapping including rock mass classification and probing where necessary.

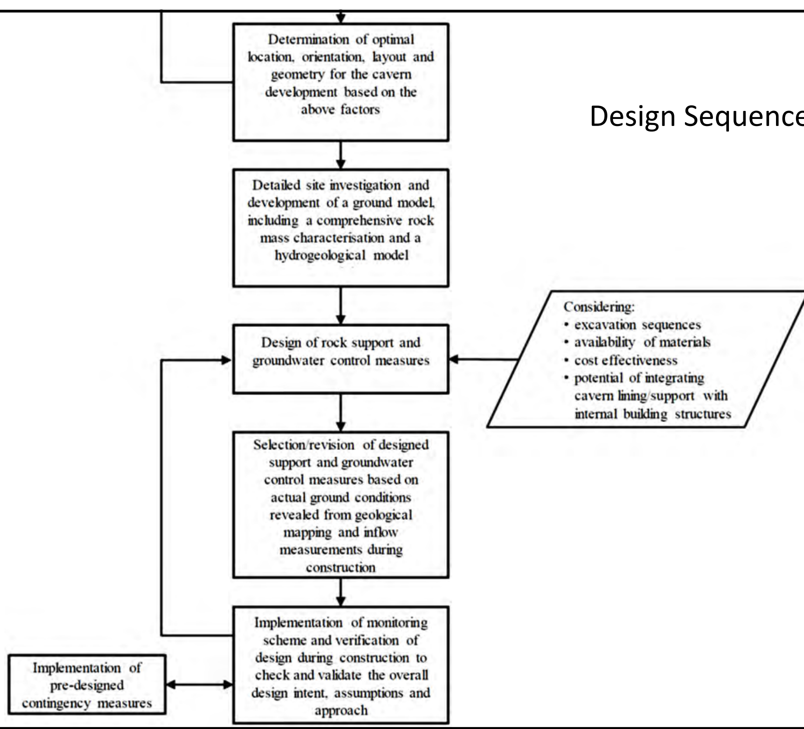
The main objective of site investigation for a tunnel/cavern development, in common with those for most other civil works, is to provide data for the:

- assessment of the suitability of the site for tunnel/cavern development and for the site selection where more than one site is considered,
- development of the ground model (particularly the rock mass characterization and groundwater model),
- design of the cavern development including optimization of size, shape, orientation and rock support,
- assessment of construction methods, time and cost, and the implications of any geological hazards and their impacts, e.g. adverse effects on hydrogeology, and
- assessment of environmental impact due to the cavern development, e.g. groundwater, vibration, noise, etc.

Design Sequence for Tunnels/Caverns



Design Sequence for Tunnels/Caverns



Design Considerations

1. Stability

A cavern should be designed to fulfil the fundamental requirements of stability during construction and throughout its design life. Instability in tunnels/caverns, including cave-in, rockfall and failure of structural supports, can result in damage to underground facilities, entrapment, injury and death of personnel. Collapse or substantial collapse of caverns may also affect surface facilities above.

2. Serviceability

Serviceability pertains to service performance requirements, for example, tunnel/cavern deformation, amount of groundwater seepage and movements that can be tolerated by sensitive receivers affected by a tunnel/cavern construction.

Design Considerations

3. Service Life

Similar to other engineered structures, a cavern is designed and built to last for a service life during which it can be maintained in a practical and economically viable manner. Support elements should be designed to be sufficiently durable and robust to guard against local deterioration of the rock mass over time, in particular, at weakness zones.

4. Other Considerations

The construction cost and time required to build a cavern depends on site accessibility and constraints, construction method and sequence, rock mass quality, groundwater inflow control and ground support requirements, amount of over-break, potential in reusing the excavated materials, need for permanent structural lining, etc. These factors should be considered holistically in order to produce an optimum design solution.

Design Considerations

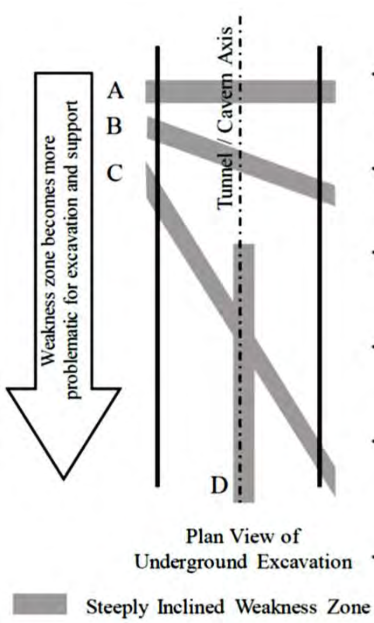
Cavern Location and Orientation

1. Factors in Optimising Cavern Location and Orientation

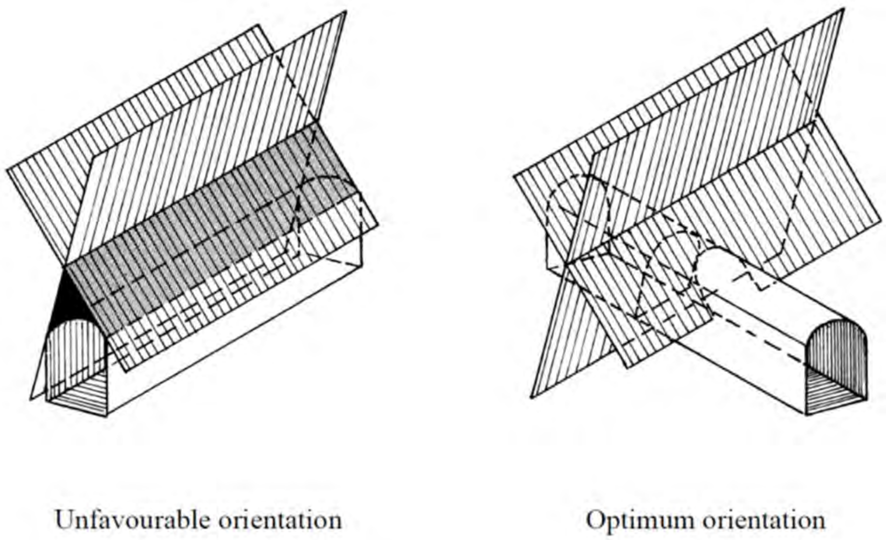
The choice of location for a cavern development is one of the most important decisions in the design process, as it determines the quality of the rock in which the cavern will be excavated.

(a) adequacy of rock cover,
(b) avoidance of rock mass weakness zones or the crossing of them in the shortest possible distance,
(c) avoidance of adverse orientations relative to major discontinuity sets,
(d) sufficient depth below groundwater table (for some uses),
(e) avoidance of rock mass under abnormally low stresses, giving reduced confinement pressures, and
(f) avoidance of rock mass under very high stresses.

Rock Support and Overbreak Related to Orientation of Weakness Zones



Case	Weakness Zone Orientation Relative to Tunnel/Cavern Axis	Sidewall Stability	Roof Stability	Overbreak
A	Perpendicular	Smaller blocks and less support needed	Localised support over short length of excavation	Likely but localised
B	Oblique (≥ 45°)	Medium blocks and moderate support needed	Moderate support over medium length of excavation	Likely and over wider section of excavation
C	Oblique (< 45°)	Larger blocks and heavier support needed	Significant support over long length of excavation	Extended with greater influence on blast hole tolerance & blockyness
D	Sub-parallel (Unfavourable)	If intersect sidewall then could destabilise major sections of sidewall, very heavy support needed	Major systematic support over very long length of excavation	Major influence on roof and sidewall overbreak & significant influence on blast hole tolerance



Orientation of Cavern with Respect to Discontinuity Sets (after Hoek &Brown, 1980)

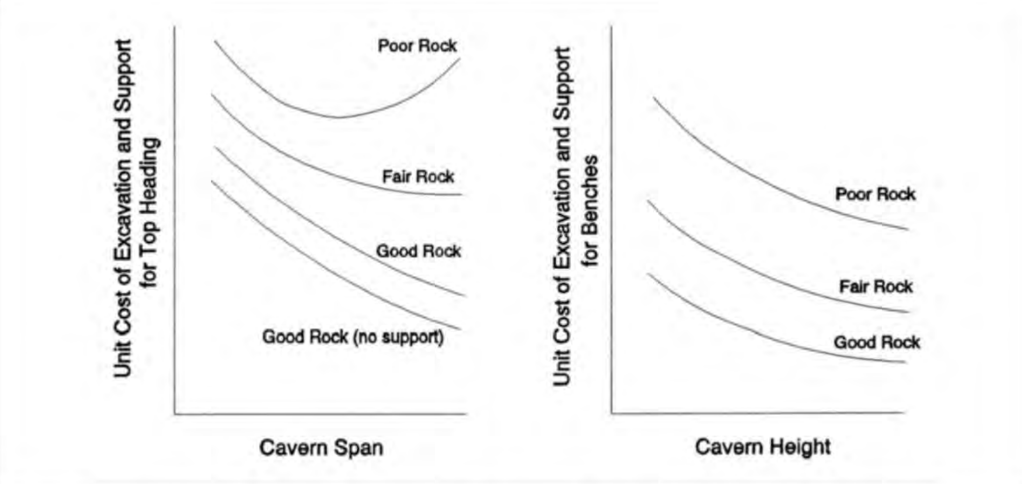
Design Considerations

Cavern Shape

A rock mass is a discontinuous material. The basic design concept for a cavern is that the shape of the opening should conform to the rock structures and stress conditions. The compressive stresses in the rock mass bounding the excavation should be evenly distributed such that the span of the cavern can be self-supporting as far as possible. The best possible stress distributions are usually obtained by giving the caverns pace a simple form with an arched roof to reduce the zone of tensile stresses in the crown.

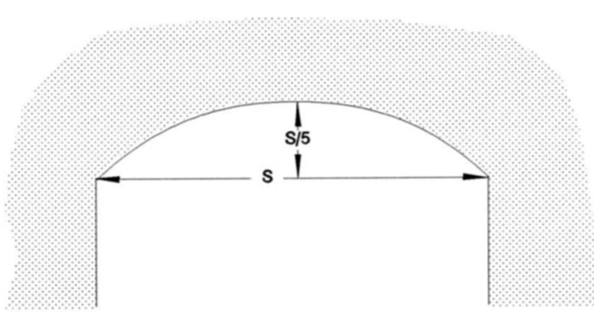
***Roof Arch*:**

The starting point for the design of the shape of a cavern roof is the assumption of a standard roof arch height of 1/5 of the cavern span. The reduction in roof support costs due to improved stability by increasing the roof arch height is normally inadequate to offset the additional excavation cost. In contrast, reducing the roof arch height increases instability problems and rockfalls potential during blasting and may only be justified if the dominant discontinuities have a shallow dip.

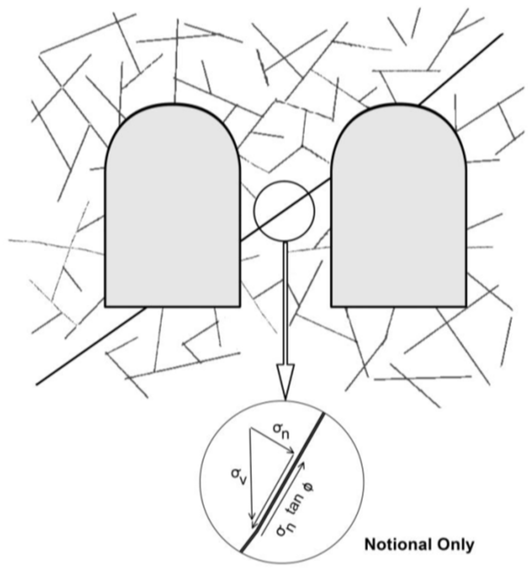


Note:	The unit cost of excavation and support refers the cost per unit volume of rock excavated.

Typical Cost Curves for Different against Cavern Span and Height for Different Rock Conditions



Standard Roof Arch Shape



Stability of Pillars with Potential Sliding Planes

Basic geological info

- Mechanical properties of the rock
- Rock type
- rock mass discontinuities
- Stress conditions
- Water conditions

Basic design stages

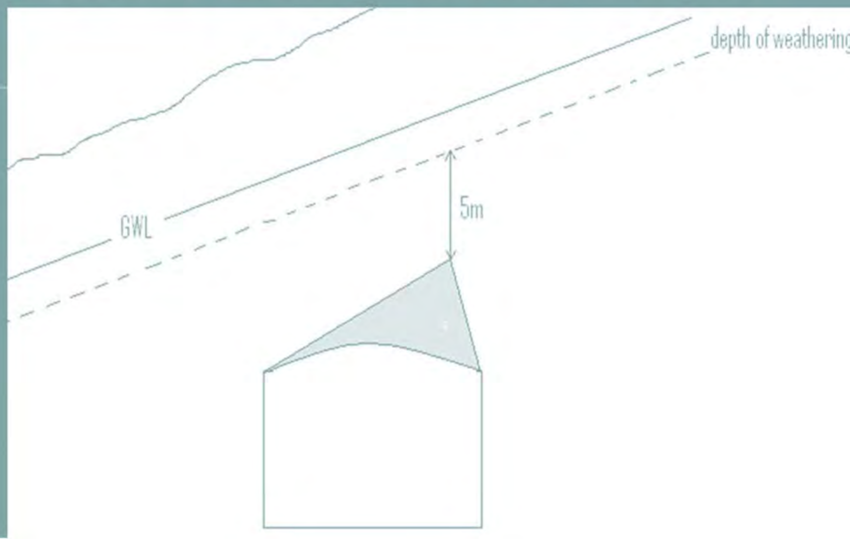
- Location of the underground opening
- Orientation of the axis
- Shaping of the opening
- Dimension of the opening

Location of the opening

- Unfavorable rock types should be avoided (e.g.. Soapstone, serpentines, periodites with clay coated joints and fissures)

- certain rocks demand may increase the economic feasibility

Necessary burden in shallow cases



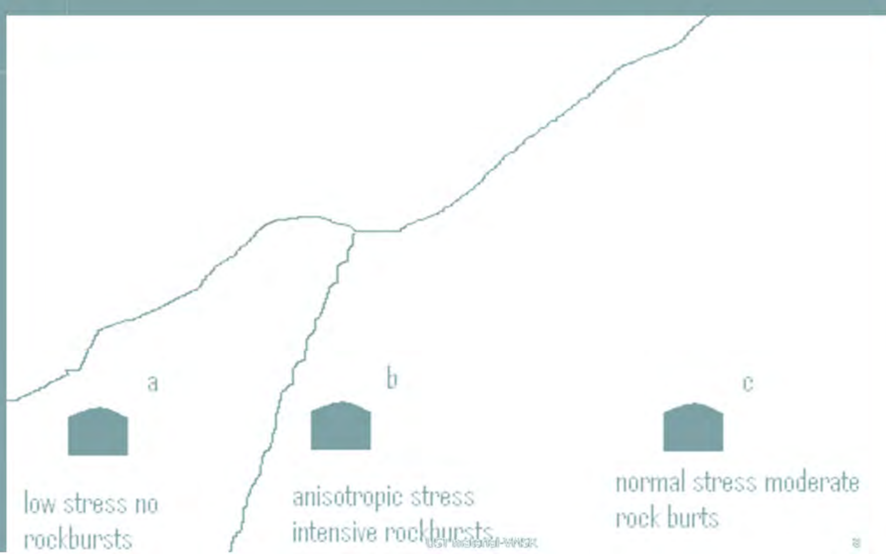
Shallow locations

- A minimum cover is required (5m for 20m span)
- Drill cores data to measure depth of weathered rock
- Borehole Permeability tests to predict the nature of joints

Deep seated locations

- Problem of high stressed rocks
- To find de-stressed zones
- safe locations
  - . protruding ‘noses’
  - . corners of valleys
  - . outside gouges, faults
- No opening should cross the faults zones

Stress situation in valley with fault zone

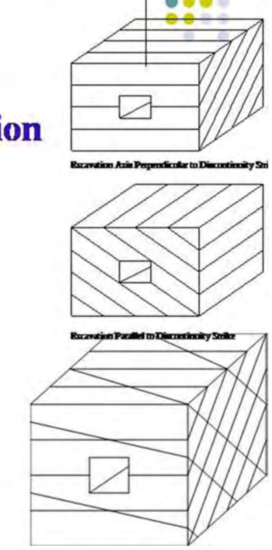


Orientation of the axis

- Strike preferably parallel to the valley
- Dip towards the valley
- Joint rosette diagram to be prepared
- Axis should be along the bisection of joint directions
- In case of deep seated axis should be at least 25 degree inclination to the smooth foliation planes

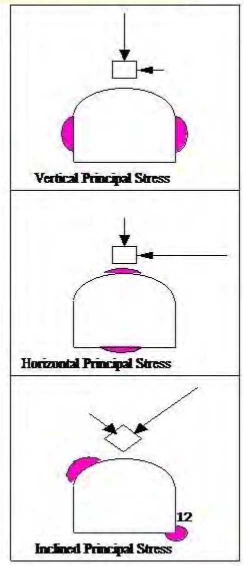
Rock Fracture - Orientation

- **Single Set of planes of weakness. Stability is a function of Excavation Axis:**
  - Maximize - Strike Perpendicular
  - Minimize - Strike Parallel
- **More typically multiple sets of planes of weaknesses..**
  - Maximize by avoiding having any strike close to parallel to axis.



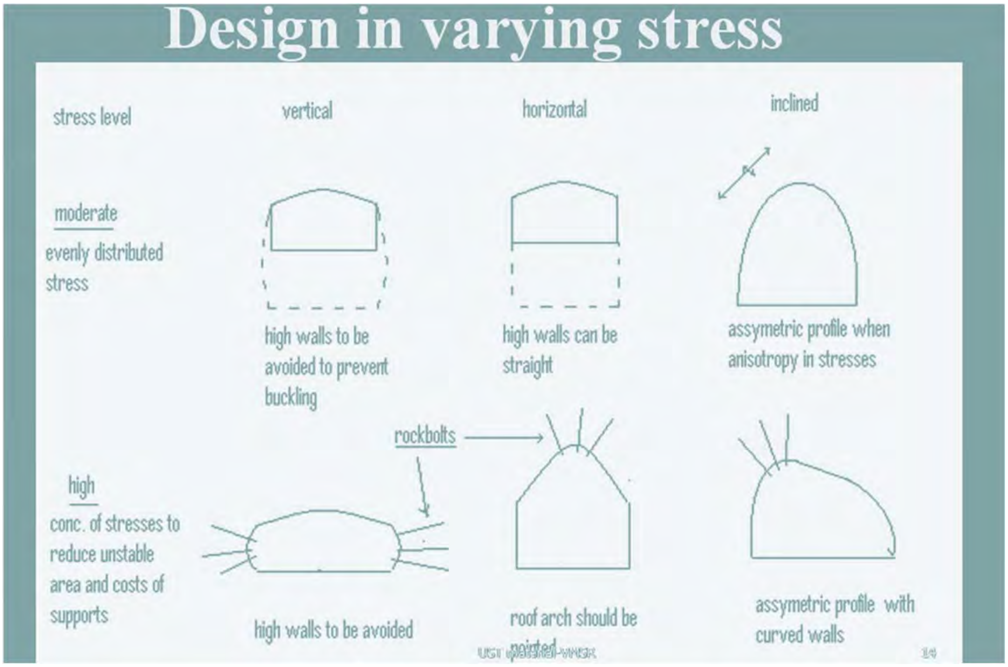
High-Stress Failure Zones

- Not always practical to have circular/elliptical sections..
- Stress concentration will occur as a function of stress field/orientation and excavation shape
- Shaded areas show where rockburst or yield is most likely to occur around a horseshoe opening under three types of principal stress orientation..
  - Vertical
  - Horizontal
  - Inclined



Shape of opening

- Generally the roof is arched
- In shallow rocks with horizontally bedded rocks flat roofs may be allowed
- in deep seated rocks spalling and failure
- small curvature radii concentrate the stresses
- In high stress conditions curvatures may be used

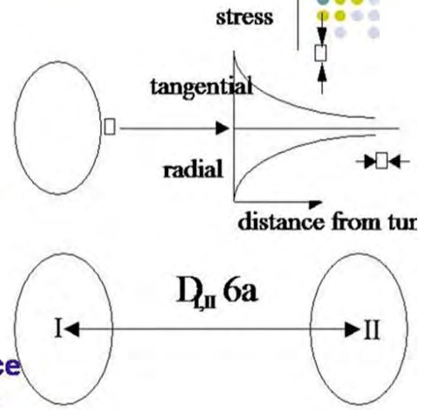


Dimension of opening

- Increasing excavation size
- Psychological factors
- Stability decreases with excavation size
- Wall thickness = ht. of rooms
- Thinner walls for high walls and vice versa

Opening Separation

- Virgin stress conditions are modified when openings are made, at the perimeter (hydrostatic stress)
  - Radial stress zero
  - Tangential stress 2x virgin
- 2 circular openings
  - Shared diameter, a
  - In hydrostatic stress field
  - Minimal Interaction if distance between openings centers is greater than 6a
- In high stress situations, ensure openings do not overly encroach on zones of influence



18

Data for Rock Mass Characterization (Sheet 1 of 2)

Rock material	• *Elastic constants (e.g. Young's modulus and Poisson's ratio)* • *Compressive strength (e.g. unconfined compressive strength (UCS), triaxial)* • *Tensile strength (e.g. Brazilian test, point load strength)* • *Density* • *Porosity* • *Anisotropy and inhomogeneity* • *Decomposition grades*
Discontinuities (e.g. weakness zones, shear fractures, joints, cleavage, etc.)	• *Orientation (dip angle / dip direction)* • *Description of sub-zones if applicable (e.g. of weakness zones)* • *Persistence, size, shape and termination* • *Spacing and frequency (including rock quality designation (RQD))* • *Block size* • *Wavelength and amplitude* • *Roughness (including joint roughness coefficient (JRC))* • *Aperture* • *Infill (including description and properties)* • *Discontinuity wall strength (e.g. joint compressive strength (JCS)) and weathering* • *Hydraulic conductivity** • *Groundwater chemistry* • *Shear strength*

Data for Rock Mass Characterization (Sheet 2 of 2)

Rock mass properties (measured using field tests or derived from rock mass classification schemes and/or appropriate failure criterion)	• *Deformability* • *Strength* • *Anisotropy and inhomogeneity* • *Weathering* • *Permeability*
In-situ Stress	• *Natural state of in-situ stress*
Excavatability	• *Brittleness value* • *Surface hardness (e.g. Vickers hardness test)* • Abrasiveness (e.g. CERCHAR) • *Indentation (e.g. punch-penetration tests)* • *Rock cutting tests (linear and rotary)* • *Drillability (Siever's J-value)* • Flakiness • *Petrographic analysis (thin section and X-ray diffraction (XRD) analysis)*

Design Considerations

Wall Height:

Cavern walls are normally vertical. This suits the method of excavation and maximizes usable space. The wall stability is a function of the wall height, the in-situ stresses and the orientation and engineering properties of the dominant discontinuity sets. A flat vertical wall surface precludes any substantial arching action and high walls tend to be unstable.

Spacing between Caverns

Typical pillar widths between caverns are between half and full cavern span or height, whichever is the greater. As a project progresses on the basis of improved geological data, the pillar widths should be optimized, and appropriate lateral confinements to the pillars should be provided if necessary.

Design for Tunnel/Cavern Structures:

Failure Modes

Failure modes in caverns depend on the characteristics of a rock mass, which is dependent on the strength of the intact rock and discontinuities, as well as the in-situ stresses. Two types of failure modes, viz. structurally controlled failure and stress-induced failure, should be considered in a tunnel/cavern design.

***Structurally Controlled Failure*:**

This type of failure involves kinematically permissible sliding on adverse joints or block falls. Examples are detachments of rock blocks, wedges or fragments, formed by intersection of pre-existing structural features, from free faces formed after excavation, including roof, sidewalls and heading face. For weakness zones or fractured rocks in discontinuities, detachment of large volume of rock fragments or partially weathered materials can also occur.

Design for Tunnel/Cavern Structures:

***Stress-induced Failure*:**

Stress-induced failures such as slabbing, rock bursting and squeezing will occur when rock materials are subject to a high external or in-situ stress. Portions of a cavern affected by stress concentration at localized areas (e.g. pillars, intersections, and areas affected by high external loads, e.g. load from foundations in the vicinity of the cavern) are also vulnerable to stress-induced failures and their stability should be duly assessed.

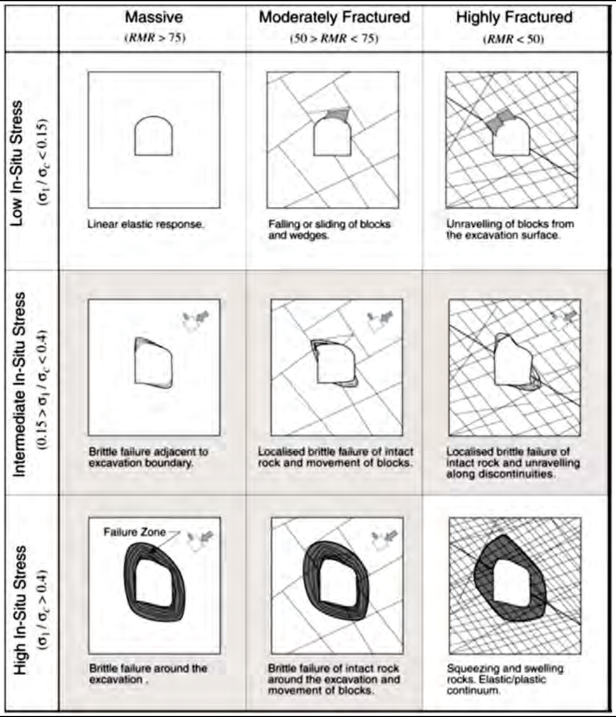


Illustration of possible failure modes according to the level of in-situ stresses and the fracture conditions of a rock mass.
(modified from Martin et al, 1999)

Design Methods and Stability of Underground Excavations (Tunnels/Caverns)

Design Methodology

Successful engineering design involves a design process, which is a sequence of events within which design develops logically. Bieniawski (1993) summarized a 10 step methodology for rock engineering design problems, incorporating 6 design principles:

STEP	DESCRIPTION	DESIGN PRINCIPLE
1	**Statement of the problem** (performance objectives)	1
2	**Functional requirements and constraints** (design variable and design issues)	1
3	**Collection of information** (site characterization, rock properties, groundwater, in situ stresses)	2
4	**Concept formulation** (geotechnical model)	3
5	**Analysis of solution components** (analytical, numerical, empirical, observational methods)	3, 4
6	**Synthesis and specifications for alternative solutions** (shapes, sizes, locations, orientations of excavations)	3, 4
7	**Evaluation** (performance assessment)	5
8	**Optimization** (performance assessment)	5
9	**Recommendation**	6
10	**Implementation** (efficient excavation, and monitoring)	6

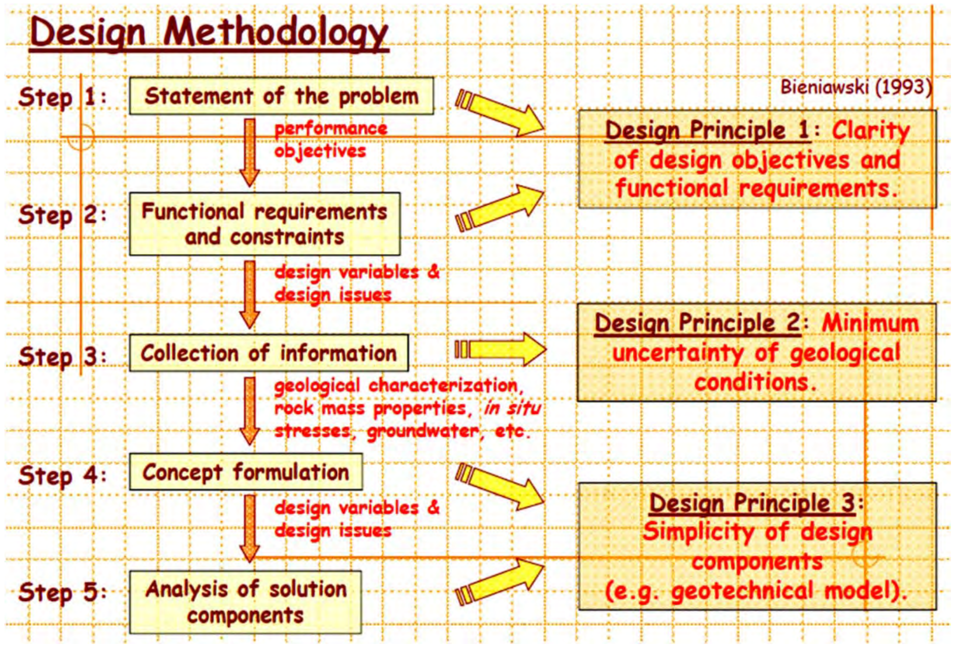
Step 1: Statement of the problem

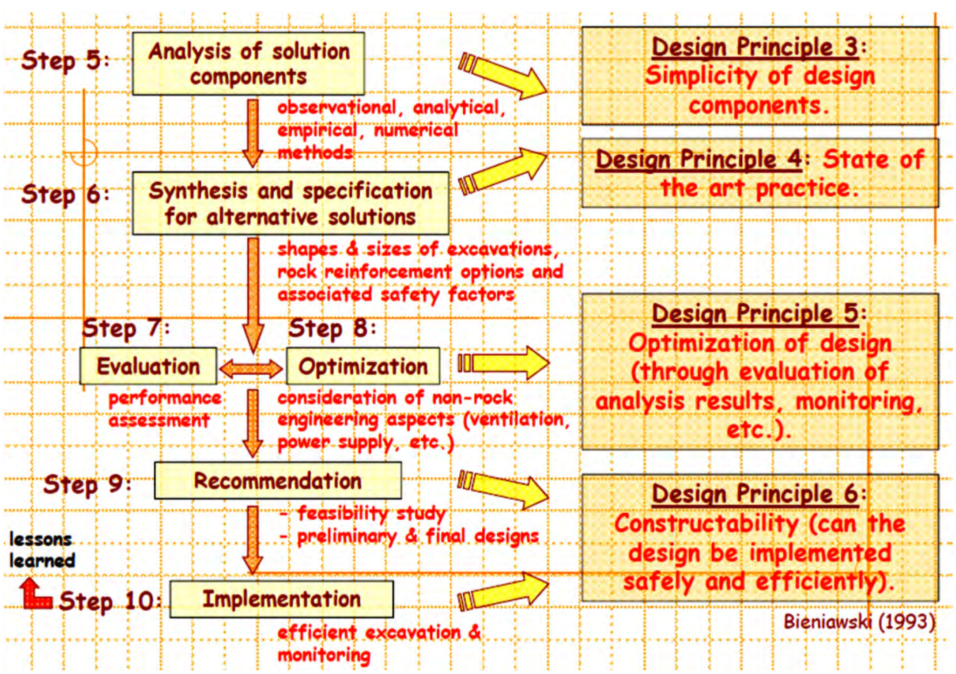
performance objectives

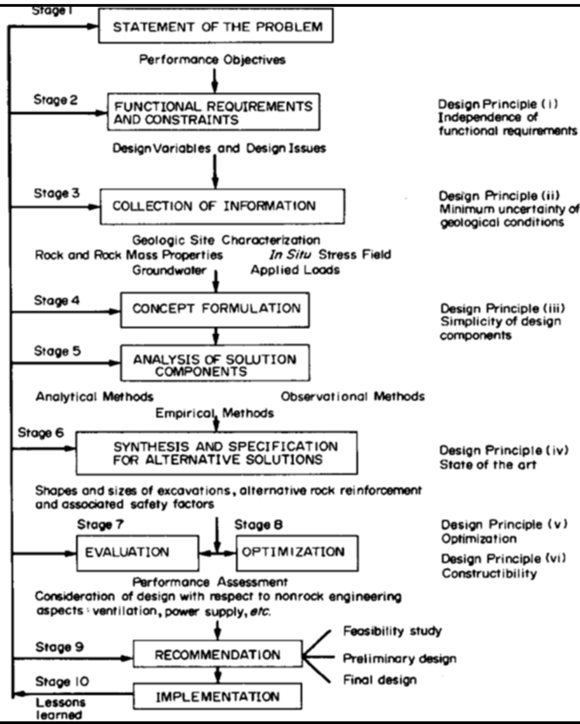
Step 2: Functional requirements and constraints

Design Principle 1: Clarity of design objectives and functional requirements.

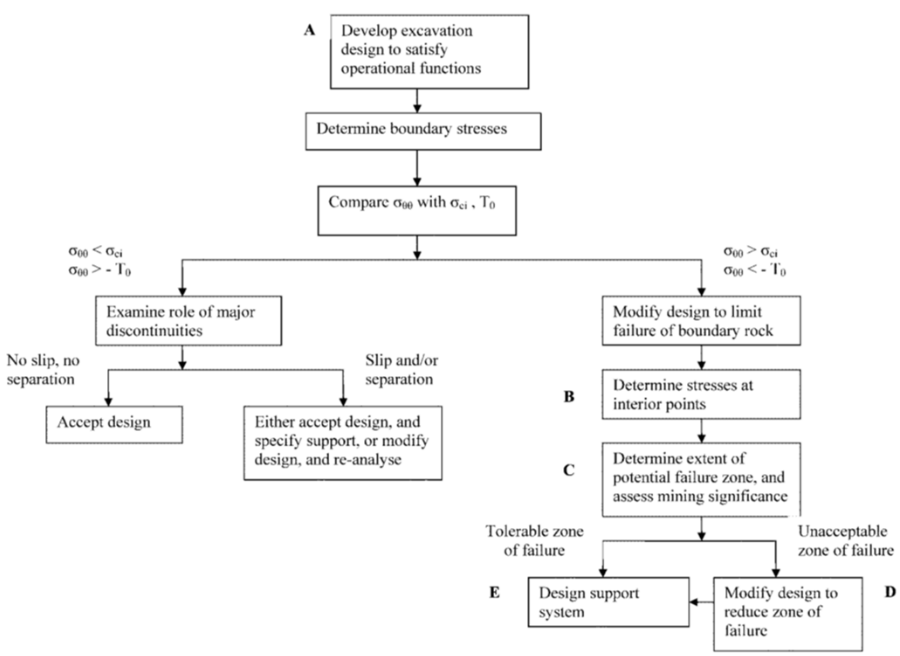
Bieniawski (1993)


Design Methodology
Bieniawski (1993)
Step 1: Statement of the problem
performance objectives
Step 2: Functional requirements and constraints
design variables & design issues
Step 3: Collection of information
geological characterization, rock mass properties, in situ stresses, groundwater, etc.
Step 4: Concept formulation
design variables & design issues
Step 5: Analysis of solution components
Design Principle 1: Clarity of design objectives and functional requirements.
Design Principle 2: Minimum uncertainty of geological conditions.
Design Principle 3: Simplicity of design components (e.g. geotechnical model).


Step 5: Analysis of solution components
observational, analytical, empirical, numerical methods
Step 6: Synthesis and specification for alternative solutions
shapes & sizes of excavations, rock reinforcement options and associated safety factors
Step 7: Evaluation
performance assessment
Step 8: Optimization
consideration of non-rock engineering aspects (ventilation, power supply, etc.)
Step 9: Recommendation
- feasibility study
- preliminary & final designs
lessons learned
Step 10: Implementation
efficient excavation & monitoring
Design Principle 3: Simplicity of design components.
Design Principle 4: State of the art practice.
Design Principle 5: Optimization of design (through evaluation of analysis results, monitoring, etc.).
Design Principle 6: Constructability (can the design be implemented safely and efficiently).
Bieniawski (1993)



A general design methodology for rock engineering (after Bieniawski, 1993)



A logical framework for underground excavation design in massive rock

DESIGN METHODS

- The design methods which are available for assessing the stability of underground rock excavations (tunnels, caverns etc.,) can be broadly categorized as:
  - Empirical methods;
  - Analytical methods;
  - Observational methods.

DESIGN METHODS

- Engineering rock mass classifications are the best-known empirical approach for assessing the stability of tunnels/caverns.
- Analytical methods involve the formulation and application of certain conceptual models for design purposes.
- The aim is to reproduce the behaviour and response of the rock excavations.

DESIGN METHODS

• Observational methods rely on actual monitoring of ground movement during excavation to detect measurable instability, and on the analysis of the observed ground-support interaction.

• This approach is the only way to verify the expected performance of excavations designed using the other two methods.

Empirical Design Methods

• They are particularly useful in the planning and preliminary design stages of a rock engineering project but, in some cases, they also serve as the main practical basis for the design of complex underground structures.

• According to Bieniawski (1989), modern rock mass classifications were developed to create some order out of the chaos in site investigation procedures and to provide the desperately needed design aids.

• Rock mass classification should be used in conjunction with observational methods and analytical studies to formulate an overall design rationale compatible with the design objectives and site geology.

Empirical Design Methods

• According to Bieniawski (1989), the objectives of rock mass classifications are:

- To identify the most significant parameters influencing the behaviour of a rock mass.
- To divide a particular rock mass formation into areas of similar behaviour, that is, rock mass classes of varying quality.
- To provide a basis for understanding the characteristics of each rock mass class.
- To relate the experience of rock conditions at one site to the conditions and experience encountered at others.
- To derive quantitative data and guidelines for engineering design.
- To provide a common basis for communication between engineers and geologists.

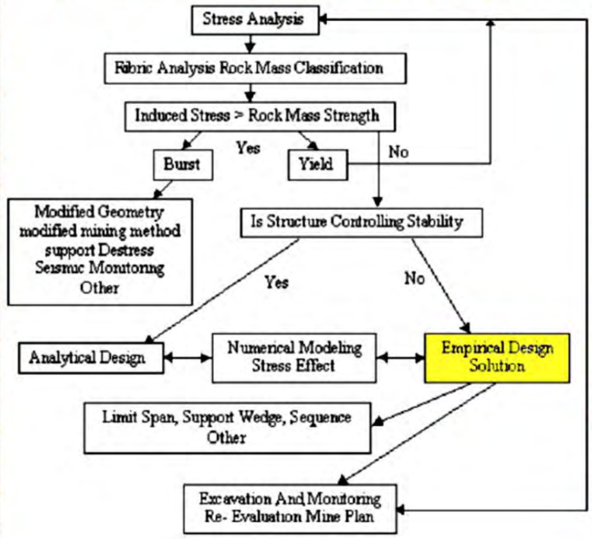
Empirical Design Methods

• All modern rock mass classification systems describe rock masses by considering various parameters together. The parameters most commonly employed are:

- rock material strength – this constitutes the strength limit of the rock mass;
- rock quality designation, RQD;
- spacing of discontinuities;
- condition of discontinuities (roughness, continuity, separation, joint wall weathering, infilling);
- orientation of discontinuities;
- groundwater conditions;
- in situ stresses.

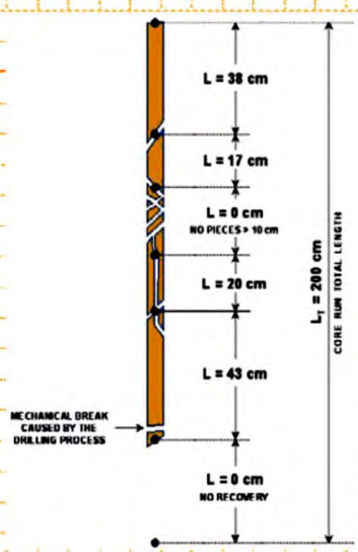
Empirical Design in Geotechnical Engineering

"Empirical design" is based upon experience or observation alone, without using scientific method or theory. Its application to engineering design relies on comparing the experiences of past practices to predict future behaviour based upon the factors most critical towards the design.



Pakalnis (1996)

Rock Quality Designation (RQD)



$$RQD = \frac{\sum Rock\ Pieces \geq 10\ cm}{Core\ Run\ Total\ Length} \times 100\,(\%)$$

$$RQD = \frac{38 + 17 + 0 + 20 + 43 + 0}{200} \times 100\,(\%)$$

$$RQD = 59\,\%\ \ (FAIR)$$

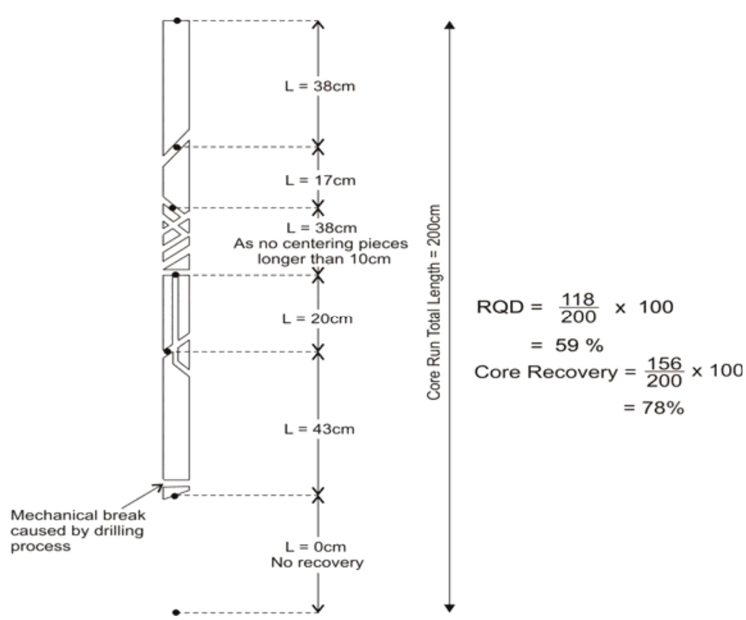
RQD (%)	Geotechnical Quality
< 25	VERY POOR
25 to 50	POOR
50 to 75	FAIR
75 to 90	GOOD
90 to 100	EXCELLENT

Correlation between RQD and rock quality

S. No.	RQD (%)	Rock Quality
1	<25	Very poor
2	25-50	Poor
3	50-75	Fair
4	75-90	Good
5	90-100	Excellent

RQD is perhaps the most commonly used method for characterizing the degree of jointing in borehole cores, although this parameter also may implicitly include other rock mass features like weathering and 'core loss' (Bieniawski, 1989).

Procedure for measurement and calculation of rock quality designation RQD (Deere, 1989)



Indirect Methods

Volumetric Joint Count

When cores are not available, RQD may be estimated from number of joints (discontinuities) per unit volume J_v.

A relationship which may be used to convert J_v into RQD for clay-free rock masses is (Palmstrom, 1982),

$$RQD = 115 – 3.3\ J_v$$

where J_v represents the total number of joints per cubic meter or the volumetric joint count. Palmstrom (2005) has proposed a new equation,

$$RQD = 110 – 2.5\ J_v$$

New correlation probably gives a more appropriate average correlation than the existing Eqn., which may be representative for the more long or flat blocks, while new Eqn. is better for blocks of cubical (bar) shape (Palmstrom, 2005).

Indirect Methods

Volumetric Joint Count

The volumetric joint count J_v has been described by Palmstrom (1982). It is a measure for the number of joints within a unit volume of rock mass defined by

$$J_v = \sum_{i=1}^{J} \left(\frac{1}{S_i}\right)$$

where S_i is the average joint spacing in metres for the i^{th} joint set and J is the total number of joint sets except the random joint set.

- Though the RQD is a simple and inexpensive index, when considered alone it is not sufficient to provide an adequate description of a rock mass because it disregards joint orientation, joint condition, type of joint filling and stress condition.

APPLICATIONS OF RQD

- Rock quality designations (RQD) has been extensively used in engineering classifications of the rock mass.

- RQD has also been used to estimate the deformation modulus of the rock mass. Zhang and Einstein (2004) have studied wider range of rock masses having RQD value ranging from 0 to 100 % and proposed the following mean correlation between RQD and modulus ratio,

$$\frac{E_d}{E_r} = 10^{0.0186\,RQD - 1.91}$$

where E_d and E_r are the deformation moduli of the rock mass and the intact rock material respectively.

Rock Mass Classification: RMR

The Rock Mass Rating (RMR) system was developed in 1973 in South Africa by Prof. Z.T. Bieniawski. The advantage of his system was that only a few basic parameters relating to the geometry and mechanical conditions of the rock mass were required.

In applying this system, the rock mass is divided into a number of structural domains and each is classified separately. Because parameters are not equally important, weighted ratings are allocated.

A. CLASSIFICATION PARAMETERS AND THEIR RATINGS

Parameter			Ranges of values						
1	Strength of intact rock material	Point-load strength index (MPa)	>10	4 - 10	2 - 4	1 - 2	For this low range, uniaxial compressive test is preferred		
		Uniaxial compressive strength (MPa)	>250	100 - 250	50 - 100	25 - 50	5 - 25	1 - 5	<1
		Rating	15	12	7	4	2	1	0
2	Drill core quality RQD (%)		90 - 100	75 - 90	50 - 75	25 - 50	<25		
		Rating	20	17	13	8	3		
3	Spacing of discontinuities		>2m	0.6 - 2m	200 - 600mm	60 - 200mm	<60mm		
		Rating	20	15	10	8	5		
4	Condition of discontinuities		Very rough surfaces Not continuous No separation Unweathered wall rock	Slightly rough surfaces Separation <1mm Slightly weathered wall rock	Slightly rough surfaces Separation <1mm Highly weathered wall rock	Slickensided surfaces or Gouge <5mm thick or Separation 1 - 5mm Continuous	Soft gouge >5mm thick or Separation >5mm Continuous		
		Rating	30	25	20	10	0		
5	Groundwater	Inflow per 10m tunnel length (l/min)	None	<10	10 - 25	25 - 125	>125		
		ratio (joint water pressure)/(major principal stress)	0	<0.1	0.1 - 0.2	0.2 - 0.5	>0.5		
		General conditions	Completely dry	Damp	Wet	Dripping	Flowing		
		Rating	15	10	7	4	0		

B. RATING ADJUSTMENT FOR DISCONTINUITY ORIENTATIONS

Strike and dip orientations		Very favourable	Favourable	Fair	Unfavourable	Very Unfavourable
Ratings	Tunnels & mines	0	-2	-5	-10	-12
	Foundations	0	-2	-7	-15	-25
	Slopes	0	-5	-25	-50	

Bieniawski (1989)

Rock Mass Classification - RMR

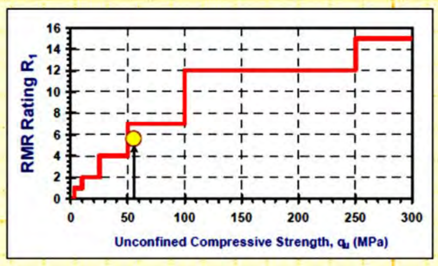
The adjusted value gives the final RMR value for the rock mass, for which several rock mass classes are described.

C. ROCK MASS CLASSES DETERMINED FROM TOTAL RATINGS					
Rating	100 ← 81	80 ← 61	60 ← 41	40 ← 21	< 21
Class number	I	II	III	IV	V
Description	Very good rock	Good rock	Fair rock	Poor rock	Very poor rock

For example:

A mudstone outcrop contains three fracture sets. Set '1' comprises bedding planes; these are highly weathered, slightly rough and continuous. The other two sets are jointing; both are slightly weathered and slightly rough. The strength of the intact rock is estimated to be 55 MPa with an RQD of 60% and a mean fracture spacing of 0.4 m. The fractures are observed to be damp.

RMR = $6+R_2+R_3+R_4+R_5$



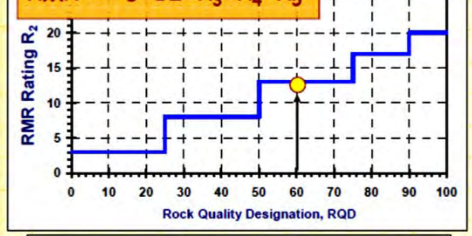
Harrison & Hudson (2000)

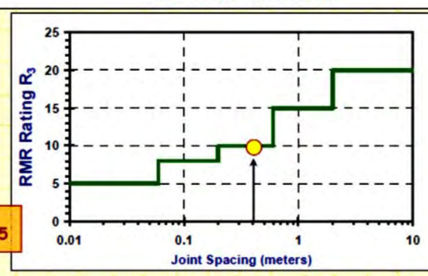
Rock Mass Classification - RMR

Example:

A mudstone outcrop contains three fracture sets. Set '1' comprises bedding planes; these are highly weathered, slightly rough and continuous. The other two sets are jointing; both are slightly weathered and slightly rough. The strength of the intact rock is estimated to be 55 MPa with an RQD of 60% and a mean fracture spacing of 0.4 m. The fractures are observed to be damp.

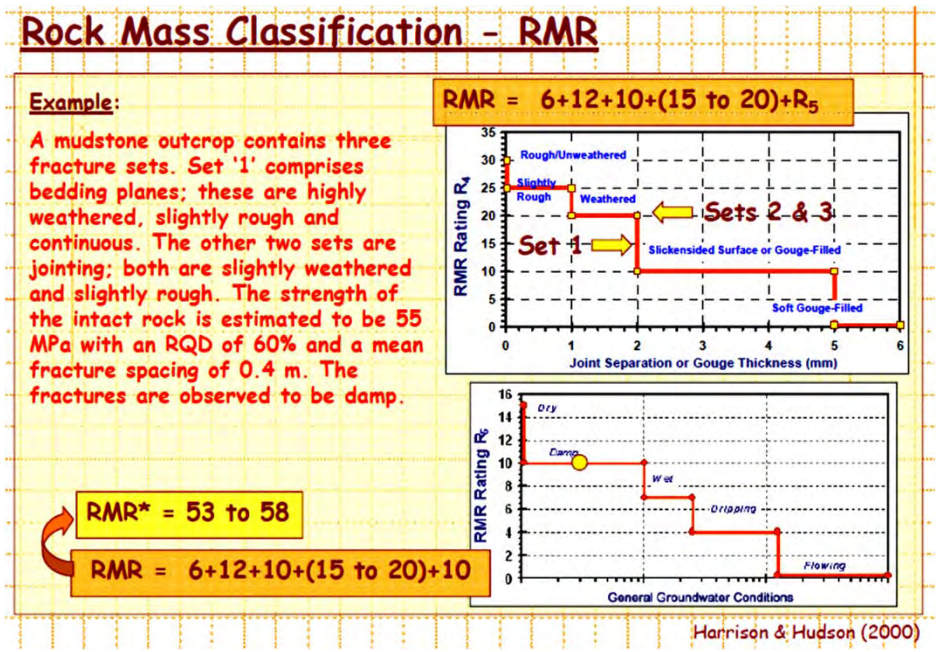
RMR = $6+12+R_3+R_4+R_5$





RMR = $6+12+10+R_4+R_5$

Harrison & Hudson (2000)



APPLICATIONS OF RMR

- Average stand-up time for arched roof
- Cohesion and angle of internal friction
- Modulus of deformation
- Estimation of support pressure

APPLICATIONS OF RMR

Average stand-up time for arched roof

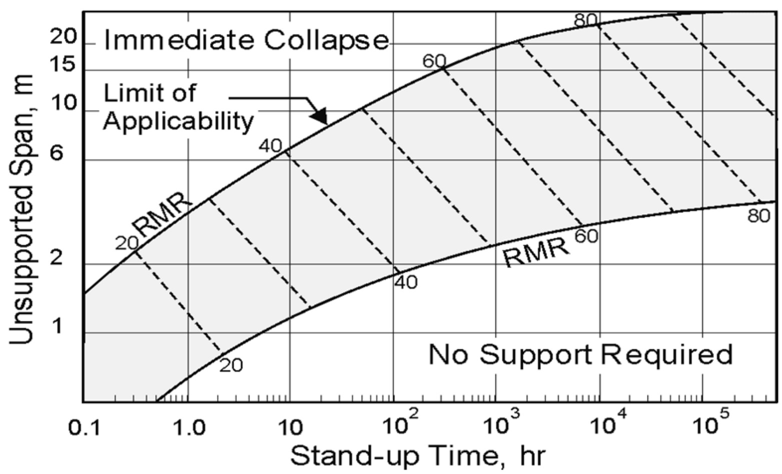
- The stand-up time depends upon effective (undisturbed) span of the opening which is defined as the width of the opening or the distance between the tunnel face and the last support, whichever is smaller.
- For arched openings the stand-up time would be significantly higher than that for a flat roof. Controlled blasting will further increase the stand-up time as damage to the rock mass is decreased.

APPLICATIONS OF RMR

Average stand-up time for arched roof

- For the tunnels with arched roof the stand-up time is related with the rock mass class.
- It is important that one should not unnecessarily delay supporting the roof in the case of a rock mass with high stand-up time as this may lead to deterioration in the rock mass which ultimately reduces the stand-up time.

Stand-up time vs. unsupported span for various rock mass classes according to RMR (Bieniawski, 1984)



Guidelines for excavation and support of rock tunnels in accordance with the rock mass rating system (Bieniawski, 1984)

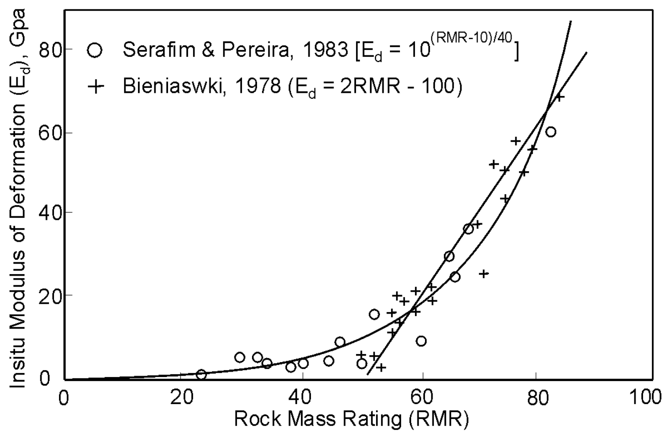
Rock Mass Class	Excavation	Supports		
		Rock bolts (20mm dia fully Grouted)	Conventional Shotcrete	Steel sets
Very good rock RMR=81-100	Full face. 3 m advance	Generally, no support required except for occasional spot bolting		
Good rock RMR = 61-80	Full face. 1.0 - 1.5m advance. Complete support 20m from face	Locally, bolts in crown 3m long, spaced 2.5m, with occasional wire mesh	50mm in crown where required	None
Fair rock RMR = 41-60	Heading and bench. 1.5 - 3m advance in head-ing. Commence support after each blast. Complete support 10m from face	Systematic bolts 4m long spaced 1.5 - 2m in crown and walls with wire mesh in crown	50-100 mm in crown and 30mm in sides	None
Poor rock RMR = 21-40	Top heading and bench. 1.0-1.5 m advance in top heading. Install support concurrently with excavation 10m from face	Systematic bolts 4-5 m long , spaced 1 - 1.5 m in crown and wall with wire mesh	100-150 mm in crown and 100 mm in sides	Light to medium ribs spaced 1.5 m where required
Very poor rock RMR <20	Multiple drifts 0.5 - 1.5 m advance in top heading. Install support concurrently with excavation. Shotcrete as soon as possible after blasting	Systematic bolts 5 -6 m long spaced 1-1.5 m in crown and walls with wire mesh. Bolt invert	150-200 mm in crown 150mm in sides and 50mm on face	Medium to heavy ribs spaced 0.75m with steel lagging and forepoling if required. Close invert.

Modulus of deformation

E_d = 2 RMR - 100 , GPa (applicable for RMR > 50)

Serafim and Pereira (1983) suggested the following correlation

$E_d = 10^{(RMR-10)/40}$, GPa (applicable for RMR < 50 also)



Correlation between modulus of deformation of rock masses and RMR (Bieniawski, 1984)

Modulus of deformation

The modulus of deformation of a dry and weak rock mass (qc<100 MPa) around underground openings located at depths exceeding 50m is dependent upon confining pressure due to overburden and may be determined by the following correlation (Verman, 1993).

$E_d = 0.3\, H\alpha .\ 10^{(RMR-20)/38}$, GPa

where,

α = 0.16 to 0.30 (higher for poor rocks), and

H = depth of location under consideration below ground surface in metres.

$\geq$ 50m

Read et al. (1999) suggested the following correlation

$E_d = 0.1\ (RMR/10)^{0.3}$, GPa

- The modulus of deformation of poor rock masses with water sensitive minerals decreases significantly after saturation and with passage of time after excavation.

Estimation of support pressure

In 1983, Unal, on the basis of his studies in coal mines, proposed the following correlation for estimation of support pressure using RMR for openings with flat roof,

$$p_V = \left[\frac{100 - RMR}{100}\right] . \gamma . B$$

where,

p_v = Support pressure,

γ = Unit weight of rock, and

B = Tunnel width.

- The estimated support pressures for non-squeezing ground conditions were unsafe for small tunnels (dia. upto 6m) and oversafe for large tunnels (diameter > 9m) which implies that the size effect is over-emphasized for arched openings. This observation is logical since bending moments in a flat roof increase geometrically with the opening size unlike in an arched roof.

Estimation of support pressure

Subsequently, using the measured support pressure values from 30 instrumented Indian tunnels, Goel and Jethwa (1991) have proposed the Eqn. for estimating the short-term support pressure for arched underground openings in both squeezing and non-squeezing ground conditions in the case of tunnelling by conventional blasting method using steel rib supports (but not in the rock burst condition).

$$p_V = \frac{7.5\ B^{0.1} .\ H^{0.5} - RMR}{20\ RMR}$$

where

B = Span of opening in metres,

H = Overburden or tunnel depth in metres (Eqn. 5 applicable for H = 50 to 600m),

p_v = Short-term roof support pressure in MPa, and

RMR = Actual (disturbed) post-excavation rock mass rating just before supporting.

- Bieniawski (1984) provided guidelines for selection of tunnel supports. This is applicable to tunnels excavated with conventional drilling and blasting method. The support measures are the permanent and not the temporary or primary supports.
- It is cautioned that the RMR system is found to be unreliable in very poor rock masses. Care should therefore be exercised to apply the RMR system in such rock mass.
- Q system is more reliable for tunnelling in the weak rock masses.

Rock Mass Classification: Q-System

The Q-system of rock mass classification was developed in 1974 in Norway by Prof. N. Barton. The system was proposed on the basis of an analysis of 212 tunnel case histories from Scandinavia.

$$Q = \frac{RQD}{J_n} \cdot \frac{J_r}{J_a} \cdot \frac{J_w}{SRF}$$

where

RQD	=	rock quality designation,
J_n	=	joint set number (related to the number of discontinuity sets),
J_r	=	joint roughness number (related to the roughness of the discontinuity surfaces),
J_a	=	joint alteration number (related to the degree of alteration or weathering of the discontinuity surfaces),
J_w	=	joint water reduction number (relates to pressures and inflow rates of water within the discontinuities), and
SRF	=	stress reduction factor (related to the presence of shear zones, stress concentrations and squeezing and swelling rocks).

... the motivation of presenting the Q-value in this form is to provide some method of interpretation for the 3 constituent quotients.

Rock Mass Classification: Q-System

The first quotient is related to the rock mass geometry. Since RQD generally increases with decreasing number of discontinuity sets, the numerator and denominator of the quotient mutually reinforce one another.

$$Q = \frac{RQD}{J_n} \cdot \frac{J_r}{J_a} \cdot \frac{J_w}{SRF}$$

The second quotient relates to "inter-block shear strength" with high values representing better 'mechanical quality' of the rock mass.

$$Q = \frac{RQD}{J_n} \cdot \frac{J_r}{J_a} \cdot \frac{J_w}{SRF}$$

The third quotient is an 'environment factor' incorporating water pressures and flows, the presence of shear zones, squeezing and swelling rock and the *in situ* stress state. The quotient increases with decreasing water pressure and favourable *in sit* stress ratios.

$$Q = \frac{RQD}{J_n} \cdot \frac{J_r}{J_a} \cdot \frac{J_w}{SRF}$$

Estimation of Support Pressure

Barton et al. (1974) plotted support capacities of 200 underground openings against the rock mass quality (Q) and found the following empirical correlation for ultimate support pressure.

$$p_v = (0.2/J_r)\, Q^{-1/3}$$

$$p_h = (0.2/J_r)\, Q_w^{-1/3}$$

where,

p_v = Ultimate roof support pressure in MPa,
p_h = Ultimate wall support pressure in MPa, and
Q_w = Wall factor.

The wall factor (Q_w) is obtained after multiplying Q by a factor which depends on the magnitude of Q as given below:

Range of Q	***Wall factor Q_w***
> 10	5.0 Q
0.1 - 10	2.5 Q
< 0.1	1.0 Q

Barton et al. (1974) further suggested that if the number of joint sets is less than three, the above Eqs. are expressed as

$$p_v = \frac{0.2.\ J_n^{1/2}}{3.\ J_r}.\ Q^{-1/3}$$

$$p_h = \frac{0.2.\ J_n^{1/2}}{3.\ J_r}.\ Q_w^{-1/3}$$

Pv and Ph are in (MPa)

The Q value in dynamic condition is half of Q value in static conditions ($Q_{dyn} = Q_{static}/2$) [Barton, 2008].

Bhasin and Grimstad (1996) suggested the following correlation for predicting support pressure in tunnels through poor rock masses (say Q < 4):

$$p_v = \frac{40\,B}{J_r} \cdot Q^{-1/3}$$

Pv in KPa

where B is diameter or span of the tunnel in metres. The above Equation shows that the support pressure increases with tunnel size B in poor rock masses.

Unsupported Span

Barton et al. (1974) proposed the following equation for estimating equivalent dimension (De') of a self supporting or an unsupported tunnel.

$$De' = 2.0\ (Q^{0.4}) \quad \text{, metres}$$

where,

D_e' = equivalent dimension,

$$= \frac{\text{span, diameter or height in metres } (B_s)}{ESR},$$

Q = rock mass quality, and

ESR = excavation support ratio,

Values of excavation support ratio ESR (Barton, 2008)

Type of Excavation		ESR
A	Temporary mine openings, etc.	2 - 5
B	Permanent mine openings, water tunnels for hydro power (excluding high pressure penstocks), pilot tunnels, drifts and headings for large openings, surge chambers	1.6 - 2.0
C	Storage caverns, water treatment plants, minor road and railway tunnels, access tunnels	1.2 - 1.3
D	Power stations, major road and railway tunnels, civil defence chambers, portals, intersections	0.9-1.1
E	Underground nuclear power stations, railway stations, sports and public facilities, factories, major gas pipeline tunnels	0.5-0.8

The excavation support ratio ESR is an inverse of the factor of safety and depends on the excavation life span and function

Design of Supports

The Q value is related to tunnel support requirements with the equivalent dimensions of the excavation.

The relationship between Q and the equivalent dimension of an excavation determines the appropriate support measures as depicted in Fig.

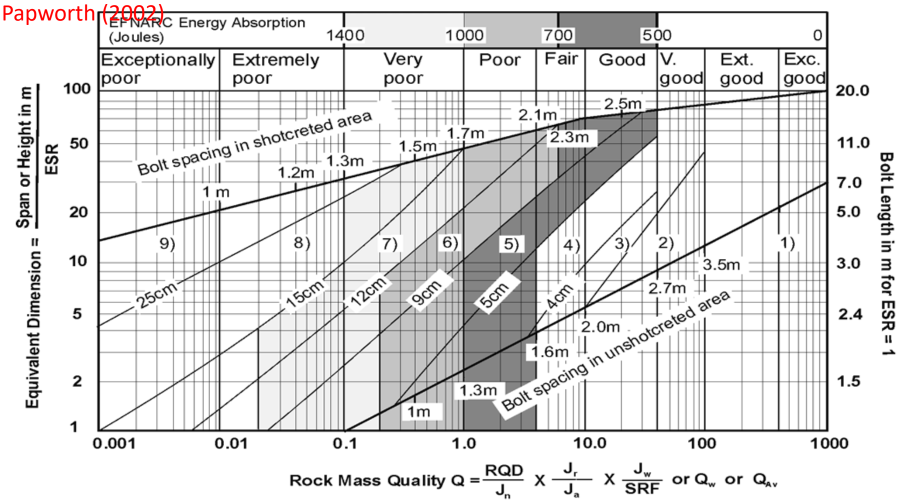
The bolt and anchor length l_b and l_a respectively, shall be determined in terms of excavation width B or height H in metres for roof and walls respectively using Eqs. proposed by Barton et al. (1974).

$$l_b = 2 + (0.15\ B \text{ or } H / ESR)\ , m$$

In Roof $l_a = 0.40\ B/ESR\ , m$

In Walls $l_a = 0.35H/ESR\ , m$

Grimstad and Barton (1993) chart for the design of support including the required energy absorption capacity of SFRS suggested by Papworth (2002)



REINFORCEMENT CATEGORIES
1) Unsupported
2) Spot bolting, sb
3) Systematic bolting, B
4) Systematic bolting (and unreinforced shotcrete, 4 to10cm, B(+S)
5) Fiber reinforced shotcrete and bolting, 5 to 9cm, S(fr)+B
6) Fiber reinforced shotcrete and bolting, 9 to 12cm, S(fr)+B
7) Fiber reinforced shotcrete and bolting, 12 to 15cm, S(fr)+B
8) Fiber reinforced shotcrete > 15cm, reinforced ribs of shotcrete and bolting, S(fr), RRS+B
9) Cast concrete lining,CCA

Other Applications

Cohesion and Angle of Internal Friction

Barton (2008) has suggested the following correlations to obtain the cohesive strength (c_p) and angle of internal friction or frictional strength (ϕ_p) of the rock mass.

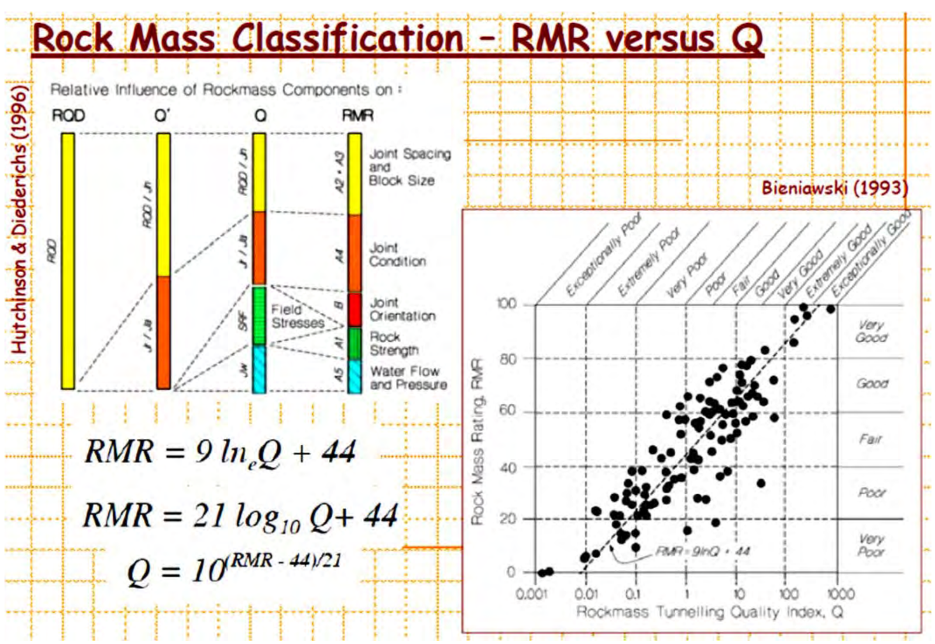
$$cp = \frac{RQD}{J_n} x \frac{1}{SRF} x \frac{q_c}{100} \quad MPa$$

$$\phi p = \tan^{-1}\left(\frac{J_r}{J_a} x J_W\right) \text{, degrees}$$

Modulus of Deformation of Rock Mass

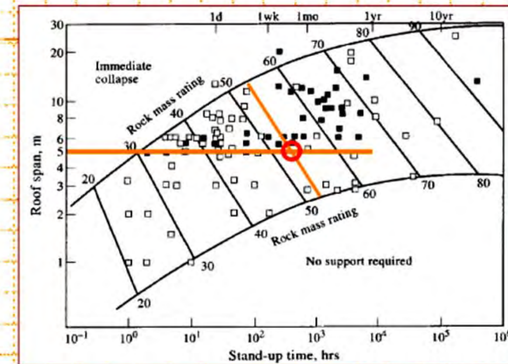
A mean value of modulus of deformation can be obtained by using the following relation (Barton, 2008).

$$E_d = 10\left(\frac{Q.q_c}{100}\right)^{1/3} \quad \text{GPa} < E_r \quad [\text{for } Q = 0.1 \text{ to } 100 \text{ and } q_c = 10\text{-}200\text{MPa}]$$



Application of Classification Systems

Rock masses can be extremely complex, making the derivation of predictive equations difficult. As an alternative, rock mass classification methods have been calibrated against large databases of case histories to provide guidelines for support design.

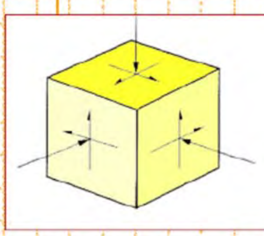


Bieniawski (1989)

Empirical design of stand-up time, the duration within which an unsupported excavation will remain serviceable, after which significant caving and failure may occur. The database used in its development examined 351 civil tunnel and underground mine case histories.

Stress-Controlled Instability Mechanisms

Structurally-controlled instabilities are generally driven by a unidirectional body force, i.e. gravity. Stress-controlled instabilities, however, are not activated by a single force, but by a tensor with six independent components. Hence, the manifestations of stress-controlled instability are more variable and complex than those of structurally-controlled failures.



$$\begin{bmatrix} \sigma_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_{zz} \end{bmatrix}$$

Stress-Controlled Instability Mechanisms

Although the fundamental complexity of the nature of stress has to be fully considered in the design of an underground excavation, the problem can be initially simplified through the assumptions of continuous, homogeneous, isotropic, linear elastic behaviour (CHILE).

CHILE: Continuous, Homogeneous, Isotropic, Linear Elastic

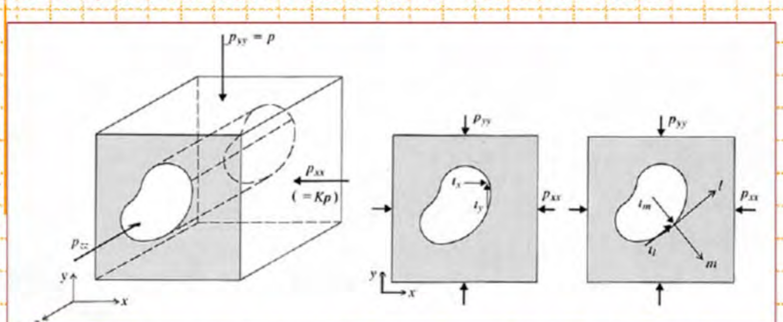
DIANE: Discontinuous, Inhomogeneous, Anisotropic, Non-Elastic

The engineering question is whether a solution based on the CHILE assumption are of any assistance in design. In fact though, many CHILE-based solutions have been used successfully, especially in those excavations at depth where high stresses have closed the fractures and the rock mass is relatively homogeneous and isotropic. However, in near-surface excavations, where the rock stresses are lower, the fractures more frequent, and the rock mass more disturbed and weathered, there is more concern about the validity of the CHILE model.

Stress-Controlled Instability Mechanisms

A stress analysis begins with a knowledge of the magnitudes and directions of the *in situ* stresses in the region of the excavation. This allows for the calculation of the excavation disturbed or induced stresses.

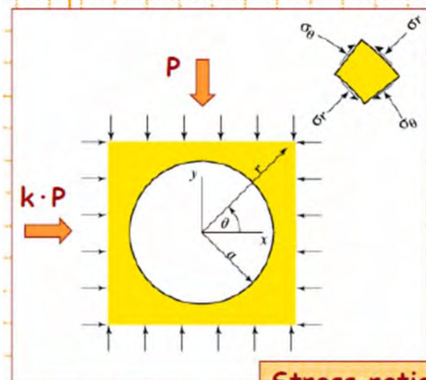
Brady & Brown (2006)



There exists several close form solutions for the induced stresses around circular and elliptical openings (and complex variable techniques extend these to many smooth, symmetrical geometries), and with numerical analysis techniques the values of the induced stresses can be determined accurately for any three-dimensional excavation geometry.

Stresses & Displacements - Circular Excavations

The Kirsch equations are a set of closed-form solutions, derived from the theory of elasticity, used to calculate the stresses and displacements around a circular excavation.



$$\sigma_{rr} = \frac{p}{2}\left[(1+K)\left(1-\frac{a^2}{r^2}\right) - (1-K)\left(1-4\frac{a^2}{r^2}+\frac{3a^4}{r^4}\right)\cos 2\theta\right]$$

$$\sigma_{\theta\theta} = \frac{p}{2}\left[(1+K)\left(1+\frac{a^2}{r^2}\right) + (1-K)\left(1+\frac{3a^4}{r^4}\right)\cos 2\theta\right]$$

$$\sigma_{r\theta} = \frac{p}{2}\left[(1-K)\left(1+\frac{2a^2}{r^2}-\frac{3a^4}{r^4}\right)\sin 2\theta\right]$$

$$u_r = -\frac{pa^2}{4Gr}\left[(1+K) - (1-K)\left\{4(1-\nu)-\frac{a^2}{r^2}\right\}\cos 2\theta\right]$$

$$u_\theta = -\frac{pa^2}{4Gr}\left[(1-K)\left\{2(1-2\nu)+\frac{a^2}{r^2}\right\}\sin 2\theta\right]$$

Brady & Brown (2006)

Stress ratio:
$k = \sigma_h/\sigma_v$

Stresses & Displacements - Circular Excavations

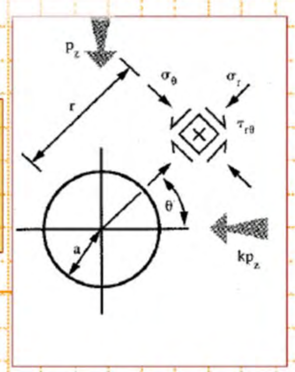
From these equations we can see that the stresses on the boundary (i.e. when $r = a$) are given by:

$$\sigma_{\theta\theta} = p[(1+k) + 2(1-k)\cos 2\theta]$$

$$\sigma_{rr} = 0$$

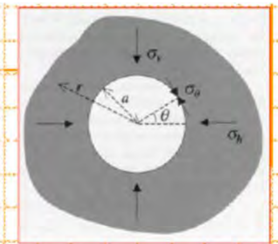
$$\tau_{r\theta} = 0$$

Note that the radial stresses are zero because there is no internal pressure, and the shear stresses must be zero at a traction-free boundary.



Q. At a depth of 750 m, a 10-m diameter circular tunnel is driven in rock having a unit weight of 26 kN/m^3 and uniaxial compressive and tensile strengths of 80.0 MPa and 3.0 MPa, respectively. Will the strength of the rock on the tunnel boundary be exceeded if:

(a) k=0.3, (b) k=2.0?



Harrison & Hudson (2000)

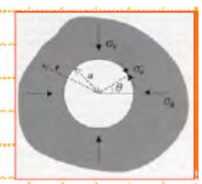
A. Since the tunnel has neither a support pressure nor an internal pressure applied to it, the local stresses at the boundary have $\sigma_3 = \sigma_r = 0$ and $\sigma_1 = \sigma_\theta$. The Kirsch solution for the circumferential stress is:

$$\sigma_\theta = \tfrac{1}{2}\sigma_v \left[(1+k)\left(1+\frac{a^2}{r^2}\right) + (1-k)\left(1+3\frac{a^4}{r^4}\right)\cos 2\theta \right]$$

For a location on the tunnel boundary (i.e. a = r), this simplifies to:

$$\sigma_\theta = \sigma_v \left[(1+k) + 2(1-k)\cos 2\theta \right]$$

Q. Circular tunnel: 750 m deep, 10m diameter, γ_{rock} = 26 kN/m^3, σ_{UCS} = 80.0 MPa, σ_T = 3.0 MPa. Will the strength of the rock on the tunnel boundary be reached if: (a) k=0.3, and (b) k=2.0?

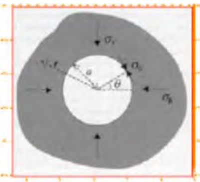


A. We assume that the vertical stress is caused by the weight of the overburden, in which case we have:

$$\sigma_v = \gamma z = 0.026 \times 750 = 19.5\ MPa$$

The extreme values of induced stress occur at positions aligned with the principal *in situ* stresses, and so in order to compute the stress induced in the crown and invert (i.e. roof and floor) we use θ = 90°, and for the sidewalls we use θ = 0° .

Q. Circular tunnel: 750 m deep, 10m diameter, γ_{rock} = 26 kN/m^3, σ_{UCS} = 80.0 MPa, σ_T = 3.0 MPa. Will the strength of the rock on the tunnel boundary be reached if: (a) k=0.3, and (b) k=2.0?



A. For k=0.3:

Crown and invert (θ = 90°), σ_θ = -1.95 MPa (i.e. tensile)
Sidewalls (θ = 0°), σ_θ = 52.7 MPa

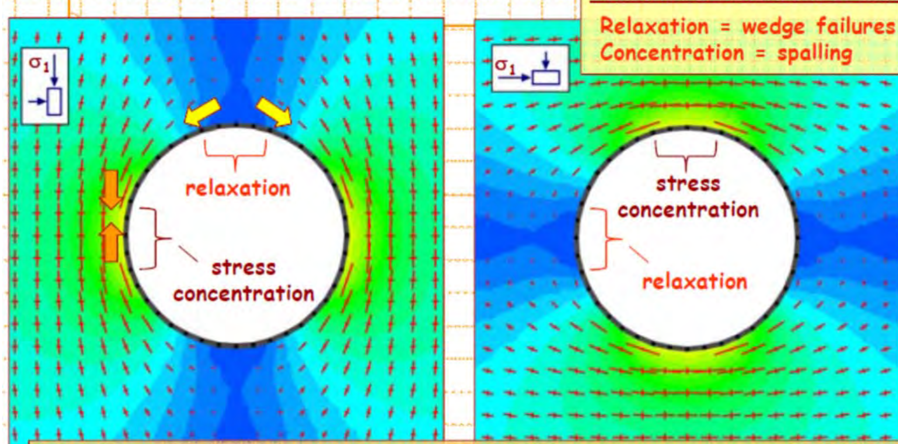
For k=2.0:

Crown and invert (θ = 90°), σ_θ = 97.5 MPa
Sidewalls (θ = 0°), σ_θ = 19.5 MPa

compressive strength is exceeded

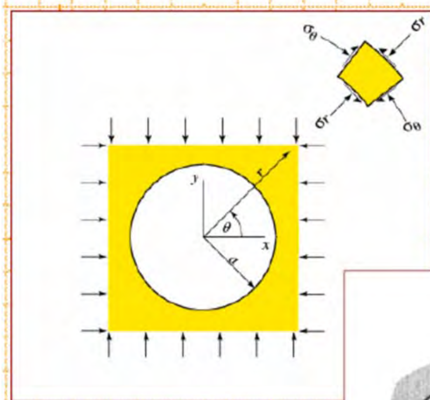
Orientation of σ_1 & Induced Stresses

Potential Ground Control Issues:
Relaxation = wedge failures
Concentration = spalling



Stresses can be visualized as flowing around the excavation periphery in the direction of the major principle stress (σ_1). Where they diverge, relaxation occurs; where they converge, stress increases occur.

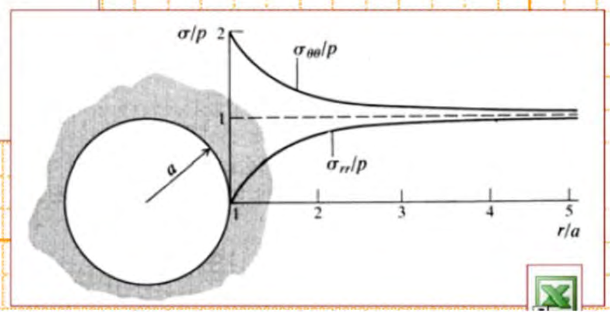
Stresses Away from Opening



$$\sigma_r = \frac{1}{2} p_z \left\{ (1+k)\left(1 - \frac{a^2}{r^2}\right) - (1-k)\left(1 - 4\frac{a^2}{r^2} + 3\frac{a^4}{r^4}\right)\cos 2\theta \right\}$$

$$\sigma_\theta = \frac{1}{2} p_z \left\{ (1+k)\left(1 + \frac{a^2}{r^2}\right) + (1-k)\left(1 + 3\frac{a^4}{r^4}\right)\cos 2\theta \right\}$$

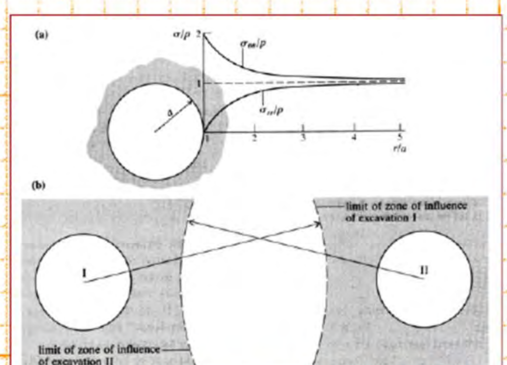
$$\tau_{r\theta} = \frac{1}{2} p_z \left\{ (1-k)\left(1 + 2\frac{a^2}{r^2} - 3\frac{a^4}{r^4}\right)\sin 2\theta \right\}$$



Zone of Influence

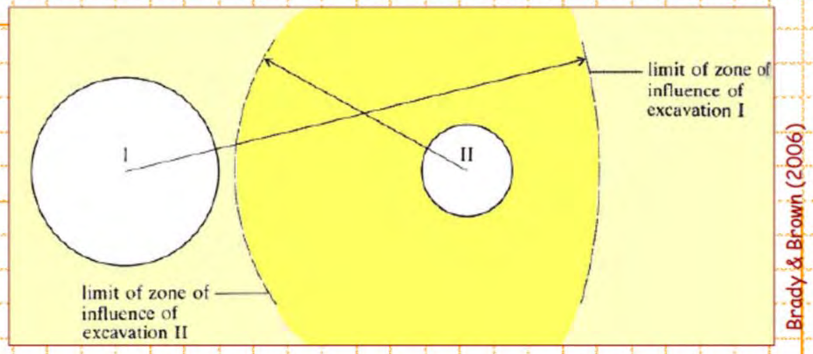
The concept of influence is important in excavation design, since the presence of a neighbouring opening may provide a significant disturbance to the near-field stresses to the point of causing failure.

... (a) axisymmetric stress distribution around a circular opening in a hydrostatic stress field; (b) circular openings in a hydrostatic stress field, effectively isolated by virtue of their exclusion from each other's zone of influence.



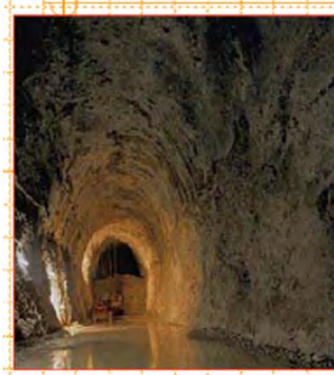
Brady & Brown (2006)

Zone of Influence



... illustration of the effect of contiguous openings of different dimensions. The zone of influence of excavation I includes excavation II, but the converse does not apply.

Stresses Around Elliptical Openings

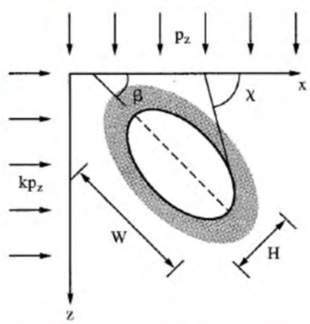


The stresses around elliptical openings can be treated in an analogous way to that just presented for circular openings. There is much greater utility associated with the solution for elliptical openings than circular openings, because these can provide a first approximation to a wide range of engineering geometries, especially openings with high width/height ratios (e.g. mine stopes, power house caverns, etc.).

From a design point of view, the effects of changing either the orientation within the stress field or the aspect ratio of such elliptical openings can be studied to optimize stability.

Stresses Around Elliptical Openings

Assuming isotropic rock conditions, an elliptical opening is completely characterized by two parameters: aspect ratio (major to minor axis) which is the eccentricity of the ellipse; and orientation with respect to the principle stresses. The position on the boundary, with reference to the x-axis, is given by the angle χ.



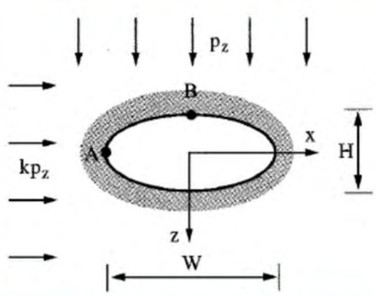
$$\sigma_\theta = \frac{p}{2q}\Big\{(1+k)\big[(1+q^2)+(1-q^2)\cos 2(\chi-\beta)\big] - (1-k)\big[(1+q)^2\cos 2\chi + (1-q^2)\cos 2\beta\big]\Big\}$$

where $q = \dfrac{W}{H}$

Hudson & Harrison (1997)

Stresses Around Elliptical Openings

It is instructive to consider the maximum and minimum values of the stress concentrations around the ellipse for the geometry of an ellipse aligned with the principal stresses. It can be easily established that the extremes of stress concentration occur at the ends of the major and minor axes.



$$\sigma_A = p(1-k+2q) = p\left(1-k+\sqrt{\frac{2W}{\rho_A}}\right),$$

$$\sigma_B = p\left(k-1+\frac{2k}{q}\right) = p\left(k-1+k\sqrt{\frac{2H}{\rho_B}}\right)$$

where, for an ellipse, the radii of curvature are

$$\rho_A = \frac{H^2}{2W} \quad \text{and} \quad \rho_B = \frac{W^2}{2H}$$

Hudson & Harrison (1997)